Software Requirements Specification

Academic Management & Record System

Tanmay T A (PES1UG22CS001) Uttam B S (PES1UG22CS002)
Vibha Vasisht (PES1UG22CS003)

Generated 2026-10-11

Contents

Revision History **2**

Approvals **3**

1 Introduction **4**
1.1 Purpose . . . 4
1.2 Scope . . . 4
1.3 Audience . . . 4
1.4 Definitions and Acronyms . . . 4

2 Overall Description **5**
2.1 Product Perspective . . . 5
2.2 Major Product Functions . . . 5
2.3 User Roles and Characteristics . . . 5
2.4 Operating Environment . . . 5
2.5 Constraints . . . 6

3 External Interface Requirements **6**
3.1 User Interfaces . . . 6
3.2 Hardware Interfaces . . . 6
3.3 Software Interfaces . . . 6
3.4 Communication Interfaces . . . 6

4 System Features (Detailed Functional Requirements) **7**
4.1 Module 1: Account Provisioning & User Profile . . . 7
4.2 Module 2: Course Setup, Self-Enrollment & Schedule Management . . . 9
4.3 Module 3: Digitized Attendance Marking & Freeze Constraints . . . 11
4.4 Module 4: Weighted Rubric Grading & Grade Lifecycle . . . 13
4.5 Module 5: Analytics & Summary Reporting . . . 15

5 Non-Functional & Security Requirements **17**
5.1 Non-Functional Requirements . . . 17
5.2 Security . . . 19

5.2.1 Security Objectives 19
5.2.2 Security Requirements 19

6 Quality Attributes & Acceptance Tests 21
6.1 Exit Criteria for Acceptance 21
6.2 Acceptance Test Suites 21

7 System Models & Diagrams 22
7.1 Architectural Model 22
7.2 Data Entity-Relationship Model 22
7.3 Grade Lifecycle State Machine 22
7.4 UML Use-Case Diagrams 22
7.4.1 Academic Operations Use-Case Diagram 22
7.4.2 System Administration Use-Case Diagram 22

8 Requirements Traceability Matrix (RTM) 22

Revision History

Version	Date	Author	Change Summary	Approval
0.1	13-09-2026	Tanmay T A	Initial SRS draft containing legacy extended requirements, initial analysis models, and appendices A–D.	Draft
0.2	27-09-2026	Uttam B S	Restructured SRS to align with standard SE template; removed out-of-scope appendices, streamlined NFR targets, eliminated complex OTP/geolocation constraints, and introduced the 48-hour attendance freeze rule.	Reviewed

Version	Date	Author	Change Summary	Approval
0.3	06-10-2026	Vibha Vasisht	Refined functional requirements: introduced student self-enrollment (max 6 courses), administrative user provisioning, restricted administrative editing of personal contact data, and defined the 4-stage grade lifecycle (`Draft` → `Submitted` → `Under Review` → `Published`).	Reviewed
0.4	09-10-2026	Tanmay T A	Finalized SMART requirement specifications (20 FRs, 5 NFRs, 2 PRJ-SO, 5 PRJ-SR), added dynamic rubric weight auto-normalization, and structured split UML Use-Case models for Academic Operations and System Administration.	Reviewed

Approvals

Role	Name	Signature / Email	Date
Course Instructor	Dr. Adithya Balasubramanyam	adithyab@pes.edu	11-10-2026

1 Introduction

1.1 Purpose

This document specifies the Software Requirements Specification (SRS) for the Academic Management & Record System. It details the functional requirements, non-functional constraints, security objectives, and system models necessary for university-level deployment. This document serves as the primary technical reference for software developers, system evaluators, and academic instructors during project implementation and acceptance testing.

1.2 Scope

The system provides digitized academic management capabilities covering administrative account provisioning, course section registration with weighted evaluation rubrics, student self-enrollment, automated schedule generation, digitized roll-call attendance marking with a strict 48-hour freeze constraint, and a multi-stage grade evaluation workflow (`Draft` → `Submitted` → `Under Review` → `Published`).

The scope explicitly excludes automated mail distribution (credentials are displayed on-screen during administrative account creation), physical hardware card integration, biometric devices, and external payment gateway processing.

1.3 Audience

This document is targeted toward: * **Software Engineering Faculty & Evaluators:** To review architecture compliance, constraint satisfaction, and acceptance criteria. * **Development Team (3 Students):** To guide API schema design, backend logic, and frontend role-based workflows during 2 one-week Agile sprints. * **System Administrators & Users:** To understand user role capabilities, system boundaries, and operational constraints.

1.4 Definitions and Acronyms

Term / Acronym	Definition
API	Application Programming Interface
GPA	Grade Point Average
JWT	JSON Web Token (used for secure session authorization)
NFR	Non-Functional Requirement
ORM	Object-Relational Mapping
RBAC	Role-Based Access Control
RTM	Requirements Traceability Matrix
SMART	Specific, Measurable, Achievable, Relevant, Time-bound criteria
SRS	Software Requirements Specification
TLS	Transport Layer Security (version 1.3)

2 Overall Description

2.1 Product Perspective

The Academic Management & Record System is an independent, responsive web service designed to replace paper-based academic tracking across a higher education institution of approximately 10,000 students and staff. It operates in a 3-tier web architecture consisting of a modern single-page web interface, a REST API application server, and a relational database store.

2.2 Major Product Functions

- **Account Provisioning & Profile Management:** Administrative creation of Student, Faculty, and Admin accounts; mandatory password changes upon first login; self-service contact detail updates.
- **Course Offering & Self-Enrollment:** Administrative course registration, assignment of primary faculty, setting evaluation rubrics (with backend auto-normalization to 100%), and student self-enrollment (capped at 6 courses per term).
- **Schedule & Roll-Call Attendance:** Class schedule publishing; digitized instructor roll-call; 48-hour automated attendance freeze; administrative override capability.
- **Weighted Rubric Grading Workflow:** Faculty mark entry against weighted components; transition through state lifecycle (`Draft` → `Submitted` → `Published`); student regrade request dispute workflow (`Under Review`).
- **Academic Dashboards:** Real-time student GPA and attendance metrics; class grade distribution summaries; administrative institution-level reporting.

2.3 User Roles and Characteristics

Role	Characteristics & Responsibilities
Student	Enrolls in up to 6 active courses during registration windows; views session-by-session attendance and grade distributions; submits regrade dispute requests; updates personal contact information.
Faculty	Conducts class roll-calls for assigned sessions; updates attendance within 48 hours; inputs rubric marks; submits draft grades for administrative review; resolves student regrade requests.
Administrator	Provisions user accounts; creates courses and schedule sessions; sets evaluation rubrics; publishes finalized grades; executes administrative attendance overrides; manages account operational states.

2.4 Operating Environment

- **Server Environment:** Linux container execution environment (Docker) running Node.js / Python application services and PostgreSQL / MySQL relational database.

- **Client Environment:** Responsive web browser (Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge) operating on devices with screen widths from 375px to 1920px.
- **Network Infrastructure:** Standard institutional WAN/LAN with mandatory HTTPS (TLS 1.3) traffic routing.

2.5 Constraints

- **Agile Sprint Scope:** Implementation is bounded by 2 one-week-long Agile sprints delivered by a 3-student engineering team.
- **Zero Mail Automation:** External SMTP server setup is excluded; temporary credentials must be rendered directly on administrative confirmation UI views.
- **Hard 48-Hour Attendance Freeze:** Faculty attendance edit permissions are strictly revoked 48 hours post-session by automated backend time checks.
- **Course Limits:** Student enrollment is capped at a strict upper limit of 6 courses per academic term.

3 External Interface Requirements

3.1 User Interfaces

- **General UI Layout:** Responsive web interface built using modern CSS layout standards. Core workflows (e.g., taking roll-call, checking GPA) must be accessible within 3 UI clicks from login.
- **Student Dashboard:** Single-screen view rendering enrolled course cards, GPA summary, aggregate attendance percentages, and grade state indicators.
- **Faculty Marking Interface:** Tabular class roster with toggle inputs for binary attendance (Present/Absent) and numerical grade input fields per rubric component.
- **Admin Control Console:** Form interfaces for provisioning user accounts, uploading class schedules, setting course rubric weights, and executing grade state updates.

3.2 Hardware Interfaces

- The system requires no direct hardware access or custom peripheral drivers. Standard client display devices, keyboards, and pointing devices (touchscreen or mouse) are supported over standard web protocols.

3.3 Software Interfaces

- **Database Management System:** Relational SQL database connected via Object-Relational Mapping (ORM) client drivers over encrypted connections.
- **Authentication Engine:** HTTP-only JSON Web Token (JWT) handling session state and role claims.

3.4 Communication Interfaces

- **Protocol:** HTTP/2 over TLS 1.3 (HTTPS) enforced across all endpoints. Plain HTTP requests automatically redirect to HTTPS with HSTS headers enabled.
- **Data Format:** Restful JSON payloads for all API requests and responses.

4 System Features (Detailed Functional Requirements)

4.1 Module 1: Account Provisioning & User Profile

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-001	The system shall allow Administrators to provision user accounts with assigned roles (Student, Faculty, Admin) and display initial temporary login credentials on a dedicated confirmation screen upon creation within 2 seconds.	Functional	High	Admin	AC-SYS-F-001: Admin creates account; temporary credentials display on screen once. Test: TC-ACC-01	Requires Admin RBAC privilege.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-002	The system shall mandate users logging in with temporary credentials to update their account password before granting access to application features.	Functional	High	Security	AC-SYS-F-002: Login with temporary password forces redirect to password change screen. Test: TC-ACC-02	Depends on SYS-F-001.
SYS-F-003	The system shall allow logged-in Students and Faculty to view and update their personal contact details (phone number, home address, emergency contact) at any time.	Functional	Medium	Student / Faculty	AC-SYS-F-003: Profile updates persist in database and display immediately upon saving. Test: TC-ACC-03	Accessible by account owner.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-004	The system shall allow Administrators to view user account metadata, reassign roles, or deactivate user accounts across the institution.	Functional	High	Admin	AC-SYS-F-004: Deactivated accounts are blocked from logging in within 1 second. Test: TC-ACC-04	Admin privilege required.

4.2 Module 2: Course Setup, Self-Enrollment & Schedule Management

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-005	The system shall allow Administrators to register course offerings, set section capacity limits, assign primary faculty, and set weighted rubric evaluation schemes with backend auto-normalization to 100%.	Functional	High	Admin / Faculty	AC-SYS-F-005: Course created with assigned faculty; raw rubric weights normalize to 100% automatically. Test: TC-CRS-01	Baseline for grading.
SYS-F-006	The system shall allow Students to self-enroll in up to 6 active course sections during an open enrollment window set by the Administrator.	Functional	High	Student	AC-SYS-F-006: Student can enroll up to 6 courses; 7th attempt or full section attempt is rejected. Test: TC-CRS-02	Requires active registration window.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-007	The system shall allow Administrators to create class schedule sessions defining course section, date, start/end time, room, and assigned instructor.	Functional	High	Admin	AC-SYS-F-007: Created schedule populates calendar views for instructor and enrolled students. Test: TC-CRS-03	Requires registered course sections.
SYS-F-008	The system shall restrict Faculty roster visibility and session access to only those course sections assigned to them by the Administrator.	Functional	High	Faculty	AC-SYS-F-008: Faculty dashboard displays only assigned sections; unauthorized access returns HTTP 403. Test: TC-CRS-04	RBAC enforcement.

4.3 Module 3: Digitized Attendance Marking & Freeze Constraints

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-009	The system shall allow Faculty to mark binary attendance (Present/Absent) for enrolled students during scheduled sessions assigned to them.	Functional	High	Faculty	AC-SYS-F-009: Attendance submission saves entries for enrolled students in valid assigned session. Test: TC-ATT-01	Depends on SYS-F-007, SYS-F-008.
SYS-F-010	The system shall automatically lock attendance modification rights for Faculty exactly 48 hours after a session's scheduled start time.	Functional	High	Operations	AC-SYS-F-010: Faculty edit controls disable 48h post-session; backend rejects edits post-48h. Test: TC-ATT-02	Automated timestamp validation.
SYS-F-011	The system shall allow Administrators to override and modify locked attendance records beyond the 48-hour window.	Functional	Medium	Admin	AC-SYS-F-011: Admin updates locked attendance record with administrative audit record. Test: TC-ATT-03	Admin privilege required.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-012	The system shall display aggregate and session-by-session attendance percentages to Students for all currently enrolled courses.	Functional	High	Student	AC-SYS-F-012: Attendance percentages update within 1s of Faculty submission. Test: TC-ATT-04	Depends on SYS-F-009.

4.4 Module 4: Weighted Rubric Grading & Grade Lifecycle

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-013	The system shall allow Faculty to record numeric marks against predefined rubric components for enrolled students, saving initial entries in `Draft` state.	Functional	High	Faculty	AC-SYS-F-013: Marks saved in `Draft` state remain hidden from student accounts. Test: TC-GRD-01	Depends on SYS-F-005.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-014	The system shall allow Faculty to submit draft grades, calculating total course scores based on normalized rubric weights and transitioning state to `Submitted`.	Functional	High	Faculty	AC-SYS-F-014: Total score calculates accurately; grade state transitions to `Submitted`. Test: TC-GRD-02	Depends on SYS-F-013.
SYS-F-015	The system shall allow Administrators to publish `Submitted` grades, changing state to `Published` and rendering final grades visible on student dashboards.	Functional	High	Admin / Student	AC-SYS-F-015: Grade state updates to `Published`; grades display on student dashboard. Test: TC-GRD-03	Admin approval step.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-016	The system shall allow Students to submit regrade requests for `Published` grades, transitioning the grade state to `Under Review`.	Functional	Medium	Student	AC-SYS-F-016: Dispute text attaches; state updates to `Under Review` on Faculty/Admin view. Test: TC-GRD-04	Student challenge workflow.
SYS-F-017	The system shall allow Faculty or Admin to resolve `Under Review` disputes by adjusting rubric marks and re-publishing the finalized grade.	Functional	Medium	Faculty / Admin	AC-SYS-F-017: Rubric marks recalculate; state transitions back to `Published`. Test: TC-GRD-05	Resolution workflow.

4.5 Module 5: Analytics & Summary Reporting

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-018	The system shall render a consolidated academic dashboard for Students summarizing overall GPA, individual course grades, and attendance rates within 800 ms.	Functional	High	Student	AC-SYS-F-018: Student dashboard loads current GPA, grades, and attendance within 800ms. Test: TC-REP-01	Consolidated view.
SYS-F-019	The system shall generate class summary reports for Faculty displaying grade distributions and attendance averages per assigned section.	Functional	Medium	Faculty	AC-SYS-F-019: Class summary view displays section mean, median, and attendance rate. Test: TC-REP-02	Faculty analytics.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SYS-F-020	The system shall allow Administrators to export institution-level summary metrics detailing total enrollments, unsubmitted grades, and attendance rates.	Functional	Medium	Admin	AC-SYS-F-020: Admin view generates institution metrics across active courses within 2s. Test: TC-REP-03	Executive reporting.

5 Non-Functional & Security Requirements

5.1 Non-Functional Requirements

Req ID	Requirement Statement (SMART)	Category	Priority	Acceptance Criteria / Measurement
SYS-NF-001	The backend API shall process 95% of standard HTTP requests with a response time of less than 800 ms under a baseline load of 500 active sessions.	Performance	High	95th percentile latency < 800 ms verified via automated load suite. Test: TC-PERF-01

Req ID	Requirement Statement (SMART)	Category	Priority	Acceptance Criteria / Measurement
SYS-NF-002	The database shall process 200 concurrent attendance and grade submission transactions without locking failures or transaction drops.	Concurrency	High	Automated concurrency test executes 200 simultaneous POST requests with 0% failure. Test: TC-PERF-02
SYS-NF-003	The application shall maintain 99.0% operational availability during active academic term periods, excluding scheduled maintenance windows.	Availability	High	System uptime monitoring logs show $\geq 99.0\%$ monthly uptime. Test: TC-OPS-01
SYS-NF-004	The web application interface shall be fully responsive across display viewports from 375px to 1920px width, allowing key workflows within 3 clicks.	Usability	Medium	UI audit passes responsiveness tests; roll-call execution completes in ≤ 3 clicks. Test: TC-UX-01
SYS-NF-005	The software architecture shall strictly separate Presentation, Application, and Data layers with automated unit test coverage of $\geq 70\%$.	Maintainability	High	Automated test coverage tools confirm $\geq 70\%$ branch/statement coverage. Test: TC-QUAL-01

5.2 Security

5.2.1 Security Objectives

ID	Objective Title	Description
PRJ-SO-001	Data Isolation & RBAC	Guarantee complete role-based data isolation so that student academic records, grades, and personal details are strictly confidential and accessible only to authorized users.
PRJ-SO-002	Workflow State Integrity	Ensure non-repudiation and integrity of academic records by enforcing immutable attendance rules (48-hour lock) and rigid grade state transitions (`Draft` → `Submitted` → `Under Review` → `Published`).

5.2.2 Security Requirements

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
PRJ-SR-001	All network communications between client web applications and system backend API services shall enforce TLS 1.3 encryption.	Security	High	Security	AC-PRJ-SR-001: Plain HTTP requests redirect to HTTPS; weak ciphers rejected. Test: TC-SEC-01	TLS 1.3 configuration.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
PRJ-SR-002	User account passwords shall be salted and hashed using bcrypt with a minimum work factor of 12 prior to database storage.	Security	High	Security	AC-PRJ-SR-002: Passwords stored strictly as bcrypt hashes in database tables. Test: TC-SEC-02	Auth Module.
PRJ-SR-003	User sessions shall be authorized using HTTP-only JSON Web Tokens (JWT) with an automated expiration lifetime of 48 hours.	Security	High	Security	AC-PRJ-SR-003: Requests with tokens older than 48 hours return HTTP 401 Unauthorized. Test: TC-SEC-03	Auth Middleware.
PRJ-SR-004	The system shall enforce Role-Based Access Control (RBAC) middleware on every API endpoint to restrict resource access according to user role.	Security	High	Security	AC-PRJ-SR-004: Accessing unauthorized role endpoints returns HTTP 403 Forbidden. Test: TC-SEC-04	RBAC Middleware.

Req ID	Requirement Statement (SMART)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
PRJ-SR-005	All user inputs shall be sanitized and parameterized using Object-Relational Mapping (ORM) queries to prevent SQL Injection and XSS attacks.	Security	High	Security	AC-PRJ-SR-005: Security vulnerability scans confirm 0 SQLi and XSS flaws on input forms. Test: TC-SEC-05	Input Sanitization.

6 Quality Attributes & Acceptance Tests

6.1 Exit Criteria for Acceptance

System acceptance for deployment requires meeting the following criteria: 1. **Functional Completeness:** 100% of high-priority Functional Requirements (`SYS-F-001` through `SYS-F-020`) implemented and verified against acceptance criteria. 2. **NFR Compliance:** Performance response time ($\leq$ 800 ms at 95th percentile) and concurrency (200 simultaneous transactions) validated through automated test scripts. 3. **Security Standards:** Zero high or critical security vulnerabilities detected; pass criteria met for all security requirements (`PRJ-SR-001` to `PRJ-SR-005`). 4. **Traceability Matrix:** 100% test case coverage verified across all requirements in the Requirements Traceability Matrix.

6.2 Acceptance Test Suites

Test Suite ID	Target Module / Area	Primary Test Focus	Target Criteria
TS-AUTH	Account Provisioning	User creation, temporary credential display, mandatory password reset.	TC-ACC-01 to TC-ACC-04 Pass
TS-CRS	Course & Schedule	Registration, student self-enrollment (max 6), roster visibility.	TC-CRS-01 to TC-CRS-04 Pass
TS-ATT	Attendance marking	Roll-call input, 48-hour automated lock, admin override.	TC-ATT-01 to TC-ATT-04 Pass

Test Suite ID	Target Module / Area	Primary Test Focus	Target Criteria
TS-GRD	Grade Lifecycle	Rubric calculation, `Draft` → `Published` states, regrade dispute.	TC-GRD-01 to TC-GRD-05 Pass
TS-PERF	System Performance	Latency $\leq$ 800 ms, 200 concurrent submission transactions.	TC-PERF-01, TC-PERF-02 Pass
TS-SEC	System Security	TLS enforcement, bcrypt hashing, JWT 48h timeout, RBAC check.	TC-SEC-01 to TC-SEC-05 Pass

7 System Models & Diagrams

7.1 Architectural Model

The application follows a standard 3-tier architecture separating user interface presentation, server-side business logic, and database persistence.

7.2 Data Entity-Relationship Model

The entity-relationship model defines user accounts, course offerings, class schedules, roll-call attendance logs, and weighted rubric grades.

7.3 Grade Lifecycle State Machine

Grades progress through a controlled state machine ensuring visibility restrictions and dispute tracking.

7.4 UML Use-Case Diagrams

7.4.1 Academic Operations Use-Case Diagram

Covers primary operational interactions executed by Students and Faculty.

7.4.2 System Administration Use-Case Diagram

Covers administrative provisioning, course setup, schedule creation, and grade publication.

8 Requirements Traceability Matrix (RTM)

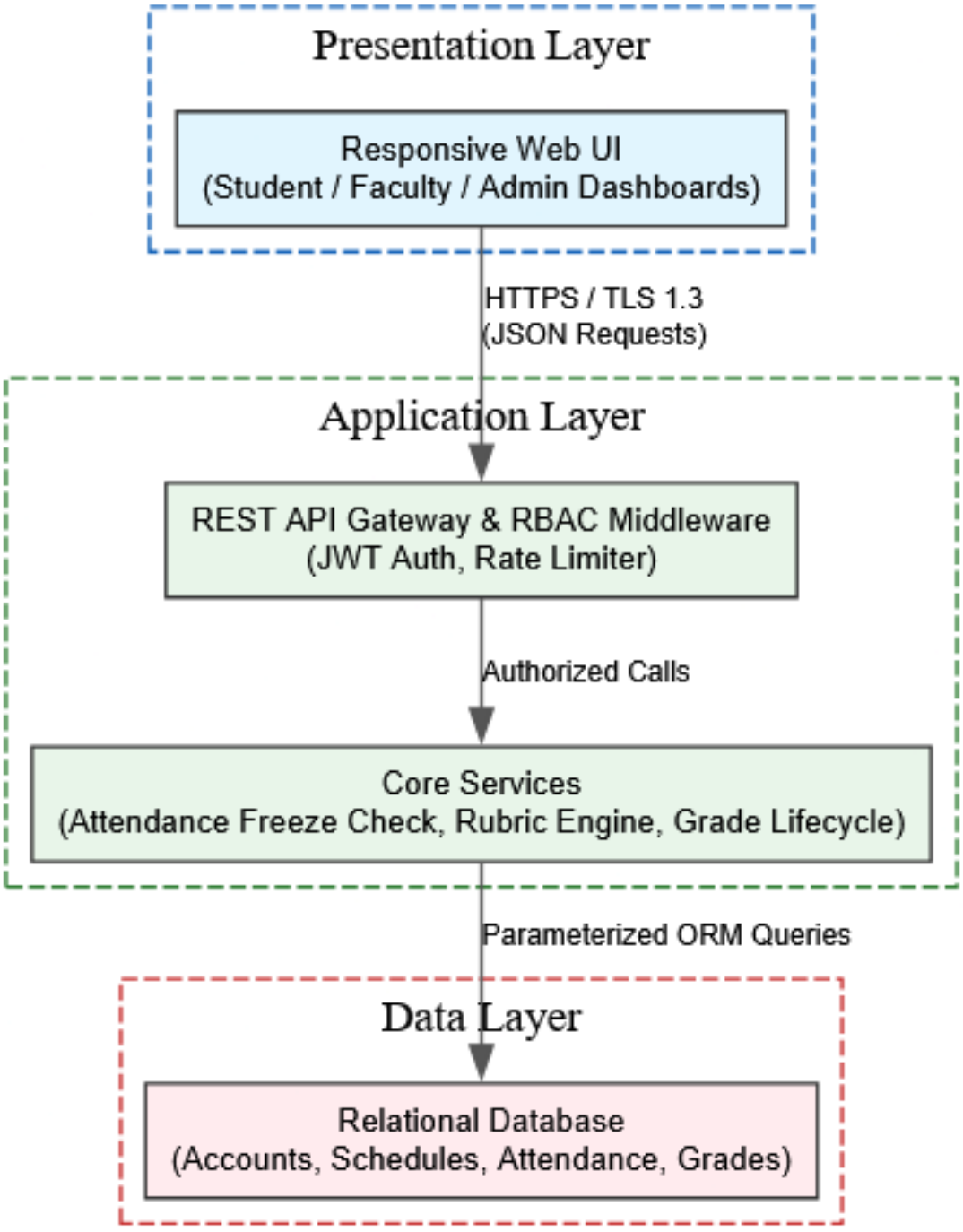


Figure 1: Three-Tier Architecture Model

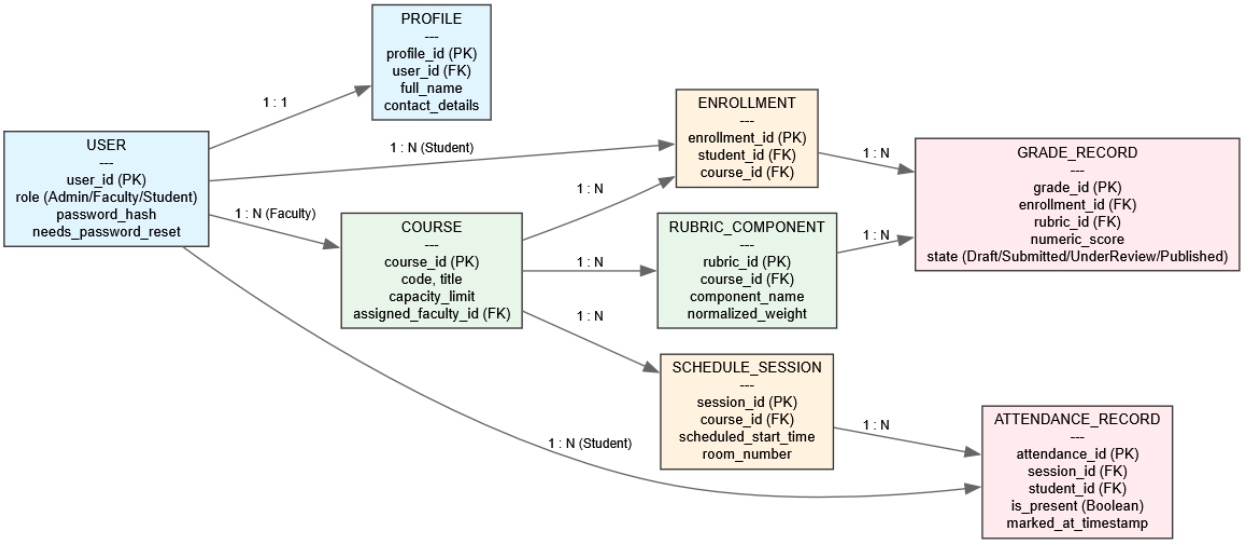


Figure 2: Entity Relationship Diagram



Figure 3: Grade Lifecycle State Machine

Req ID	Requirement Summary	Design Spec Ref	Target Module	Test Case Ref	Status
SYS-F-001	Provision user accounts & display credentials	DS-ACC-01	Account Module	TC-ACC-01	Planned
SYS-F-002	Mandate password update on first login	DS-ACC-02	Auth Module	TC-ACC-02	Planned
SYS-F-003	Update personal contact details	DS-ACC-03	Profile Module	TC-ACC-03	Planned
SYS-F-004	Manage account metadata and status	DS-ACC-04	Admin Module	TC-ACC-04	Planned
SYS-F-005	Register course & set rubric scheme	DS-CRS-01	Course Module	TC-CRS-01	Planned

Req ID	Requirement Summary	Design Spec Ref	Target Module	Test Case Ref	Status
SYS-F-006	Student self-enrollment (max 6 courses)	DS-CRS-02	Enrollment Module	TC-CRS-02	Planned
SYS-F-007	Create and publish class schedules	DS-CRS-03	Schedule Module	TC-CRS-03	Planned
SYS-F-008	Restrict faculty roster visibility	DS-CRS-04	RBAC / Course	TC-CRS-04	Planned
SYS-F-009	Digitized binary roll-call attendance	DS-ATT-01	Attendance Module	TC-ATT-01	Planned
SYS-F-010	48-hour automated attendance freeze	DS-ATT-02	Attendance Module	TC-ATT-02	Planned
SYS-F-011	Administrative attendance override	DS-ATT-03	Admin / Attendance	TC-ATT-03	Planned
SYS-F-012	Student attendance dashboard view	DS-ATT-04	Dashboard Module	TC-ATT-04	Planned
SYS-F-013	Record rubric marks in `Draft` state	DS-GRD-01	Grading Module	TC-GRD-01	Planned
SYS-F-014	Submit draft grades with weight total	DS-GRD-02	Grading Module	TC-GRD-02	Planned
SYS-F-015	Publish final grades to students	DS-GRD-03	Admin / Grading	TC-GRD-03	Planned
SYS-F-016	Submit regrade dispute (`Under Review`)	DS-GRD-04	Dispute Module	TC-GRD-04	Planned
SYS-F-017	Resolve dispute and re-publish grade	DS-GRD-05	Grading Module	TC-GRD-05	Planned

Req ID	Requirement Summary	Design Spec Ref	Target Module	Test Case Ref	Status
SYS-F-018	Student consolidated GPA dashboard	DS-REP-01	Dashboard Module	TC-REP-01	Planned
SYS-F-019	Faculty class summary analytics	DS-REP-02	Analytics Module	TC-REP-02	Planned
SYS-F-020	Admin institution-level metrics export	DS-REP-03	Analytics Module	TC-REP-03	Planned
SYS-NF-001	Latency < 800 ms at 95th percentile	DS-PERF-01	System-wide	TC-PERF-01	Planned
SYS-NF-002	200 concurrent transaction capability	DS-PERF-02	Database Layer	TC-PERF-02	Planned
SYS-NF-003	99.0% operational availability	DS-OPS-01	Infrastructure	TC-OPS-01	Planned
SYS-NF-004	Responsive UI (375px-1920px, ≤ 3 clicks)	DS-UX-01	Web Presentation	TC-UX-01	Planned
SYS-NF-005	3-tier architecture & $\geq 70\%$ test cover	DS-QUAL-01	Architecture	TC-QUAL-01	Planned
PRJ-SO-001	Data isolation and RBAC objective	DS-SEC-01	System-wide	TC-SEC-04	Planned
PRJ-SO-002	Workflow state integrity objective	DS-SEC-02	Core Business Rules	TC-ATT-02, TC-GRD-03	Planned
PRJ-SR-001	Mandatory TLS 1.3 encryption	DS-SEC-03	Network Layer	TC-SEC-01	Planned
PRJ-SR-002	Bcrypt password hashing (factor 12)	DS-SEC-04	Auth Module	TC-SEC-02	Planned

Req ID	Requirement Summary	Design Spec Ref	Target Module	Test Case Ref	Status
PRJ-SR-003	48-hour HTTP-only JWT sessions	DS-SEC-05	Auth Middleware	TC-SEC-03	Planned
PRJ-SR-004	Endpoint RBAC enforcement middleware	DS-SEC-06	API Middleware	TC-SEC-04	Planned
PRJ-SR-005	ORM parameterization (SQLi/XSS prevention)	DS-SEC-07	Data Access Layer	TC-SEC-05	Planned

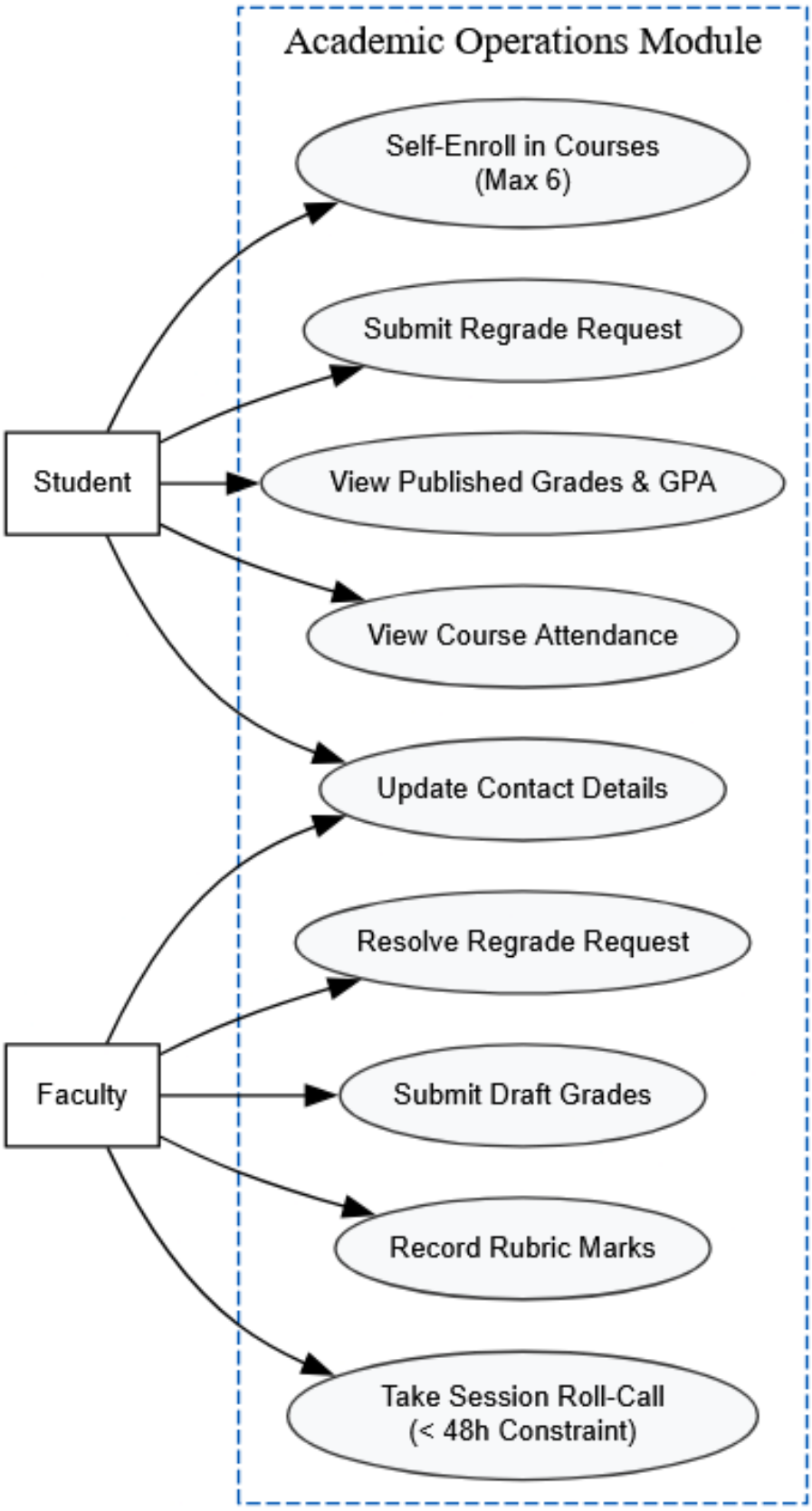


Figure 4: Academic Operations Use Case Diagram

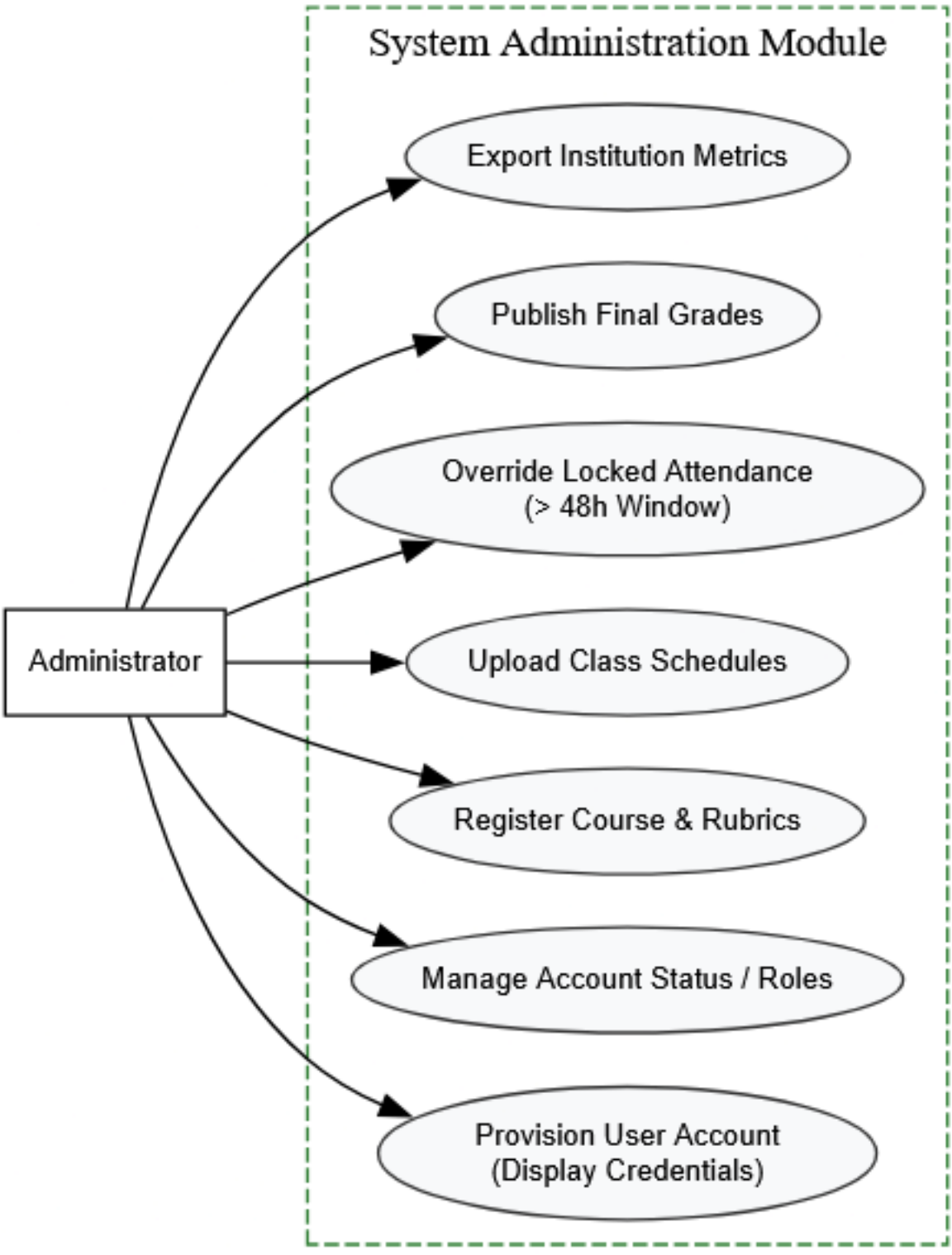


Figure 5: System Administration Use Case Diagram